Assimilation of SAR Ice and Open Water Retrievals in Environment and Climate Change Canada Regional Ice-Ocean Prediction System

Alexander S. Komarov, *Member, IEEE*, Alain Caya, Mark Buehner, and Lynn Pogson

***Abstract*— In this article, we evaluate the impact of assimilating spaceborne synthetic aperture radar (SAR) data in an Arctic regional ice analysis system over a year cycle. Ice and water information was automatically extracted from more than 7000 RADARSAT-2 HH-HV ScanSAR Wide images acquired over the Canadian Arctic and adjacent waters throughout the entire year 2013. A quality-control procedure was specifically developed and applied to reduce the number of erroneous SAR retrievals. To assess the impact of SAR ice and water retrievals on the Environment and Climate Change Canada (ECCC) Regional Ice-Ocean Prediction System (RIOPS) ice concentration analyses, we designed a set of data assimilation experiments with and without the inclusion of SAR retrievals. Our verification results suggest that the assimilation of SAR-derived retrievals considerably improves ice concentration analyses in the situations where high spatial resolution is important (e.g., near land and over small inland lakes). Furthermore, SAR retrievals are particularly useful over the areas where the Canadian Ice Service's (CIS) manually derived ice products (such as Image Analyses, daily and weekly ice charts) are not available or have limited coverage. The three-satellite RADARSAT Constellation Mission (RCM) launched in June 2019 will significantly increase the temporal frequency of SAR data. According to the most recent CIS estimate, more than 54 000 RCM images a year will be acquired over the CIS areas of interest. Therefore, the assimilation of SAR retrievals from RCM should further enhance automated ice concentration analyses products.**

***Index Terms*— Data assimilation, ice concentration analysis, RADARSAT-2, Regional Ice-Ocean Prediction System (RIOPS), synthetic aperture radar (SAR).**

I. Introduction

ARCTIC sea ice numerical prediction systems play an increasingly important role in studying physical, chemical, and biological processes in the rapidly changing Arctic marine environment [1]. Various operational activities in the Arctic Ocean such as shipping and natural resource extraction often rely on sea ice model forecasts. Numerical weather prediction models require accurate information on Arctic sea ice, since it can have a significant influence on the midlatitude atmospheric conditions [2], [3].

Performance of a regional sea ice model significantly depends on the model's initial state. To accurately specify initial conditions of a sea ice model, various types of spaceborne remote sensing observations are typically used. The ice concentration analysis component of Regional Ice-Ocean Prediction System (RIOPS) developed and maintained operationally at Environment and Climate Change Canada (ECCC) assimilates several sources of remote sensing data to produce the spatial distribution of Arctic ice concentration [4]–[6]. The RIOPS domain covers the entire Arctic with an average resolution of 5 km. Currently, passive microwave Special Sensor Microwave Imager (SSM/I), Special Sensor Microwave Imager Sounder (SSMIS), Advanced Microwave Scanning Radiometer 2 (AMSR2), ASCAT scatterometer, optical AVHRR data, as well as the Canadian Ice Service's (CIS) manually derived ice products [7] are assimilated in RIOPS. Each observation set has limitations. The passive microwave and active scatterometer observations have relatively low spatial resolution (∼25–50 km), which significantly reduces their usefulness in the vicinity of coasts and over small inland lakes. The optical data are heavily dependent on cloud cover and solar illumination, and the temporal and spatial coverage of the manually produced ice charts are limited.

Spaceborne synthetic aperture radar (SAR) observations with very high spatial resolution (at or below 100 m) have been extensively used to routinely monitor sea ice conditions as they have large coverage (up to 500 km × 500 km) and are not dependent on clouds and solar illumination. Many spaceborne SAR platforms are currently operational, including Canada's RADARSAT-2 (C-band), Europe's Sentinel-1A and 1B (C-band), TanDEM-X (X-band), and Japan's PALSAR-2 (L-band), and provide a large volume of SAR observations. Furthermore, the Canadian three-satellite RADARSAT Constellation Mission (RCM) [8] that was successfully launched in June 2019 will drastically increase the amount of SAR observations over the Arctic in the near future. Therefore, it could be beneficial to directly assimilate high-resolution SAR data in RIOPS, which will potentially result in improved ice analyses and forecasts near the ice edge, land, and in narrow channels. However, automated retrieval of ice information from SAR is challenging owing to significant variations and possible overlap of radar backscatter from open water and

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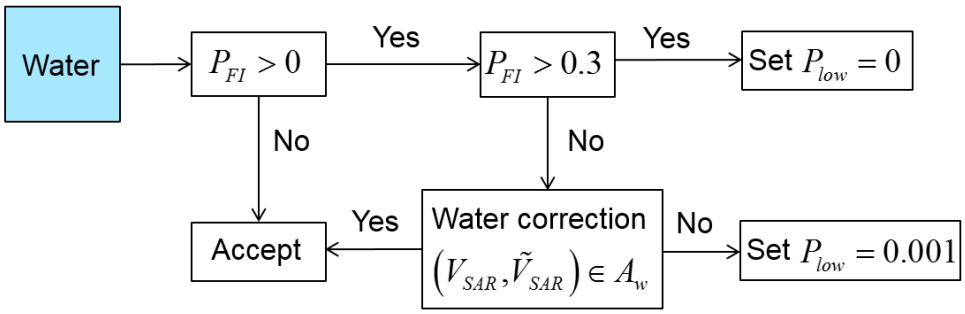


Fig. 1. Flowchart for quality-control procedure applied to water retrievals. P_{FI} is the probability of land fast ice based on the land fast ice climatology derived from NIC ice charts; V_{SAR} is the SAR wind speed; $\tilde{V}_{\mathrm{SAR}}$ is the standard deviation of the SAR wind speed; and A_w is a polygon in the space of $\{V_{\mathrm{SAR}}, \tilde{V}_{\mathrm{SAR}}\}$ as shown in Fig. 2.

ice throughout the ice season. At the same time, only SAR retrievals with very high accuracy could be considered for assimilation in RIOPS, because the ice concentration analysis system already uses multiple observation sources.

Komarov and Buehner [9] recently introduced a new approach for the automated detection of ice and open water from RADARSAT-2 HH-HV ScanSAR Wide (SCW) images. The approach calculates the probability of the presence of ice within 2.05 km × 2.05 km areas. The technique was further extended to increase the accuracy of retrievals by introducing adaptive probability thresholding depending on local SAR signatures, while preserving the large number of retrievals [10]. In this article, we automatically derive ice information from all available RADARSAT-2 HH-HV SCW images acquired by CIS over a 12-month period in 2013 and assess the impact of additionally assimilating the SAR retrievals on the RIOPS ice concentration analyses. This article presents first results for SAR data assimilation in a regional Arctic ice prediction system over a year cycle.

The objectives of this article are as follows: 1) to conduct SAR ice/water retrievals for the year 2013 using the developed approach with new quality-control procedures applied, 2) to develop and implement a series of data assimilation experiments with and without including SAR data, and 3) to assess the impact of SAR retrievals on the resulting ice concentration analyses in different regions and seasons by comparing with various ice data sets.

II. SAR Retrievals for Data Assimilation

A. SAR Data

We collected 7411 RADARSAT-2 HH-HV SCW images with 50-m pixel size [11] acquired over the Canadian Arctic and adjacent waters monitored by the CIS over the entire year 2013. The dual-polarization HH-HV SCW (approximately 500 km × 500 km) is the primary RADARSAT-2 mode for sea and lake ice monitoring at the CIS. The collected data is independent of the SAR data set used for training the ice probability model (IPM) [9] summarized below.

B. Ice and Water Detection Approach

The probability of the presence of ice within 2.05 km × 2.05 km areas (i.e., 41 × 41 pixel at 50-m resolution) at a given location in a RADARSAT-2 HH-HV SCW image is calculated according to [9] as follows:

$$P(\mathbf{x}) = (1 + \exp[f(\mathbf{x})])^{-1}, \tag{1}$$

where $f(\mathbf{x})$ is a polynomial function of three predictor variables $\mathbf{x} = \{x_1, x_2, x_3\}$. x_1 is the difference in wind speeds derived from the HH-HV SAR image using the model described in [12] and from the ECCC Global Environmental Multiscale (GEM) model regional deterministic forecasts [13]; x_2 is the spatial correlation between HH and HV backscatter over the 41 × 41 pixel window; x_3 is the standard deviation of SAR-derived wind speeds within that window. A 3 × 3 pixel median filter was applied to both HH and HV channels of the RADARSAT-2 image to reduce speckle noise prior to calculating the predictor variables.

To determine if the tested area is ice or open water, the calculated ice probability P is compared against low P_{low} and high P_{high} probability thresholds [10]. P_{low} and P_{high} are calculated as functions of SAR-derived wind speed and the probability of the presence of ice computed according to the simplified 2-D version of IPM [10]. If $P > P_{\mathrm{high}}$, then the tested area is assigned as ice; if $P < P_{\mathrm{low}}$, then the tested area is assigned as water; if $P_{\mathrm{low}} \leq P \leq P_{\mathrm{high}}$, then the tested area is assigned to be unknown, meaning that it could not be classified as ice or open water with sufficiently high confidence.

C. Quality Control

Each ice and water retrieval derived by the approach summarized above has gone through a quality-control procedure described below. The quality control largely relies on the long-term ice climatology derived from the U.S. National Ice Center (NIC) bi-weekly ice charts [14]. Various types of observations are manually interpreted and combined as part of the development of NIC charts. Currently, SAR observations include those from Sentinel-1A/B and RADARSAT-2; optical observations are provided by MODIS, Advanced Baseline Imager (ABI), AVHRR, Operational Line-Scan System (OLS), and VIIRS; passive microwave data come from AMSR2 and SSMIS, and scatterometer data are provided by ASCAT [15]. Buoy and ship data as well as model outputs provide additional supportive information.

1) Quality Control for Water Retrievals: Fig. 1 shows a flowchart of the proposed quality-control procedure applied

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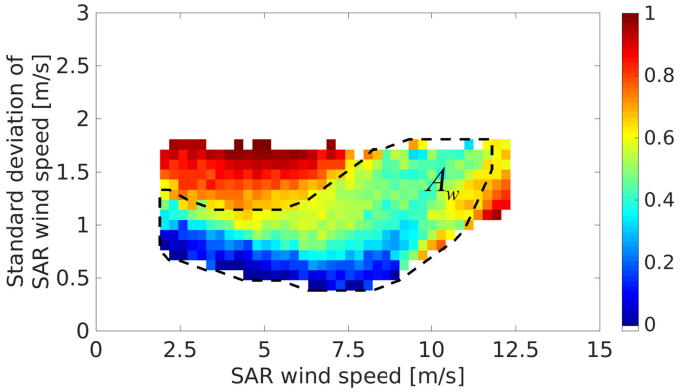


Fig. 2. Probability of having misclassified ice (i.e., ice classified as water) in the space of the SAR wind speed and standard deviation of the SAR wind speed. The number of correctly classified water samples is 3 403 309, and the number of misclassified ice samples is 17 006. A_w denotes the polygon enclosing the area of low probabilities of misclassified ice.

to water retrievals. Two key criteria are required for this procedure as illustrated in Fig. 1.

The first criterion is based on the long-term 12-years climatology of the presence of land fast ice derived from the NIC bi-weekly ice charts over the time period between 2006 and 2017. The derived climatology represents the probability of the presence of land fast ice P_{FI} over the entire Arctic domain calculated on the RIOPS grid with the average spatial resolution of 5 km and the temporal resolution of 14 days. The rationale behind considering P_{FI} is driven by the presence of systematic persistent erroneous water retrievals (i.e., ice incorrectly classified as water) over land fast ice areas mainly in the Canadian Arctic Archipelago during winter time. This is explained by the fact that the land fast ice surface in winter is very smooth which often leads to very low backscatter values similar to those obtained over calm water and, therefore, resulting in erroneous retrieval of water.

The second criterion is based on the area of low probabilities A_w of misclassified water in the space of the SAR wind speed V_{SAR} and the standard deviation of the SAR wind speed $\tilde{V}_{\mathrm{SAR}}$. Empirical probability of the presence of misclassified ice in the space of V_{SAR} and $\tilde{V}_{\mathrm{SAR}}$ is shown in Fig. 2. This probability was calculated based on the pure ice and water training subset introduced in [9] and [10]. The polygon enclosing low probabilities of misclassified ice A_i was derived using the approach described in [10].

According to Fig. 1, if the probability of land fast ice P_{FI} is higher than 0.3, then the low probability threshold P_{low} is forced to 0 to eliminate a potentially erroneous water retrieval (i.e., the water observation is rejected); if $0 < P_{\mathrm{FI}} \leq 0.3$, then the water correction procedure is applied; finally, if $P_{\mathrm{FI}} = 0$, then the water retrieval is accepted. The water correction procedure assigns the low probability threshold P_{low} to 0.001 if the current point does not belong to the A_w polygon.

2) Quality Control for Ice Retrievals: Fig. 3 shows a flowchart for the quality-control procedure applied to ice retrievals. It involves two main criteria that were derived based on our findings of systematic ice retrievals over icebergs calved from glaciers on the west Greenland coast during summer time. The icebergs with sizes of several pixels are detected as ice owing to the high spatial correlation between HH and HV, leading to high ice probability. Such retrievals are not desirable for assimilation in RIOPS, as the ice concentration within 2.05 km × 2.05 km areas that contain icebergs and labeled as ice would likely be, in reality, very low.

The first criterion is the long-term 12-year bergy water and ice-free climatology derived from NIC bi-weekly ice charts over the time period between 2006 and 2017. Bergy water and ice-free categories are specific terms that are commonly used in NIC and CIS ice charts. Bergy water corresponds to an area of freely navigable water in which ice of land origin is present; other ice types may be present, but the total ice concentration is less than 10%; ice free corresponds to an area with absolutely no ice present [7]. Probabilities of the presence of bergy water P_{BW} and ice free P_{IF} over the entire Arctic domain were calculated with the average spatial resolution of 5 km and the temporal resolution of 14 days. In the initial version of the quality-control approach, the probability of bergy water P_{BW} was derived only from NIC charts; however, this value was found to be quite low over areas known to have a high population of icebergs, which suggests that many icebergs were not taken into account in the NIC charts. We found that the ice-free category was assigned to many areas containing icebergs in NIC charts; therefore, the probability of having ice-free areas (which potentially contain an iceberg) P_{IF} was added. The ice quality-control procedure is restricted to the geographical area encompassing Baffin Bay and Greenland coast (Fig. 4) which corresponds to the area where a large number of icebergs are consistently observed.

The second criterion is based on the area of high probability A_i of having ice in bergy water or ice-free categories in the space of V_{SAR} and $\tilde{V}_{\mathrm{SAR}}$. Empirical probability of having ice in bergy water and ice-free categories in the space of V_{SAR} and $\tilde{V}_{\mathrm{SAR}}$ is shown in Fig. 5. This probability was calculated based on the training data for ice and water (ice free combined with the bergy water category) subsets introduced in [9] and [10]. The polygon enclosing high probabilities of ice in bergy water or ice-free categories, denoted by A_i, was derived using the approach described in [10].

According to Fig. 3, if the probability of bergy or ice-free water $P_{\mathrm{BW}} + P_{\mathrm{IF}}$ is higher than 0.9, then the high probability threshold P_{high} is forced to 1, which eliminates the ice retrieval; if $0 < P_{\mathrm{BW}} + P_{\mathrm{IF}} \leq 0.9$, then the ice correction procedure is applied; finally, if $P_{\mathrm{BW}} + P_{\mathrm{IF}} = 0$, then the ice retrieval is accepted. The ice correction procedure rejects the ice retrieval if the current point belongs to the A_i polygon.

D. SAR Retrieval Options

The following three types of SAR retrievals were generated using the ice and water detection approach proposed in [9] and [10] with the quality-control procedure described in Section II-C: 1) ice and water retrievals at a 2.05 km × 2.05 km scale (i.e., 41 × 41 pixel) with 5 km spacing between the centers of the retrievals: 2) ice and water retrievals at a 2.05 km × 2.05 km scale with 2.05 km spacing between

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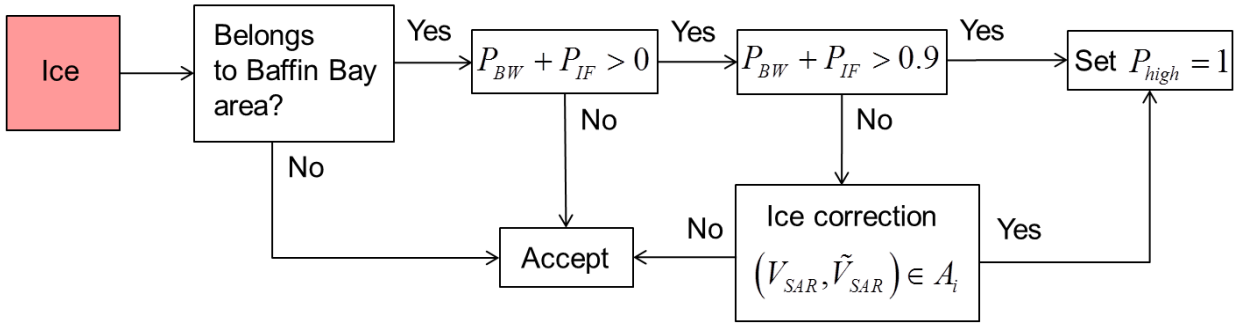


Fig. 3. Flowchart for the quality-control procedure applied to ice retrievals. P_{BW} is the probability of bergy water based on the bergy water climatology derived from NIC ice charts; P_{IF} is the probability of ice free based on the ice-free climatology derived from NIC ice charts; A_i is a polygon in the space of $\{V_{\mathrm{SAR}}, \tilde{V}_{\mathrm{SAR}}\}$ as shown in Fig. 4.



Fig. 4. Geographical area encompassing Baffin Bay and the Greenland coast where the quality-control procedure is applied to ice retrievals.

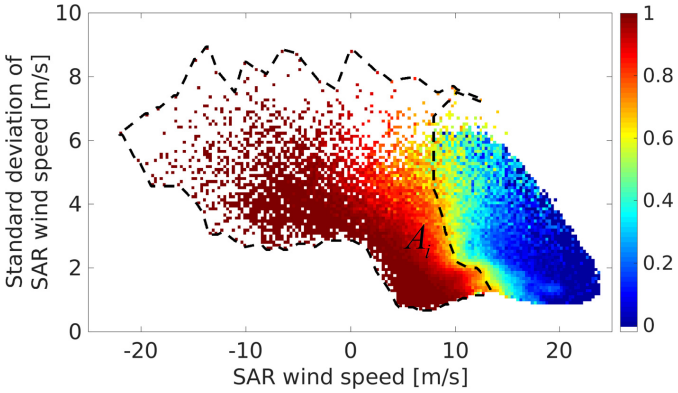


Fig. 5. Probability of having ice in bergy water or ice-free categories in the space of the SAR wind speed and standard deviation of the SAR wind speed. The number of correctly classified ice samples is 3 831 107, and the number of ice retrievals in the bergy water and ice-free categories is 96 530. A_i denotes the polygon enclosing the area of high probabilities of ice in the bergy water and ice-free categories.

the centers of the retrievals; and 3) relative number of ice retrievals within a 3 × 3 ice/water retrieval box (6.15 km × 6.15 km) with 6.15 km spacing (which could provide estimate of ice concentration in some situations). An example of the derived retrievals with these three options from the same SAR image acquired on September 22, 2013 over the Chukchi Sea is shown in Fig. 6.

III. Data Assimilation Experiments

To assess the impact of assimilating the SAR retrievals on the RIOPS ice concentration analysis, we conducted a series of data assimilation experiments for the year 2013 as summarized in Table I. In the Control experiment, all available satellite data sources (AMSR2, SSM/I, SSMIS, ASCAT, and AVHRR) are used. In the Charts experiment, CIS manually derived products (mostly from SAR images) are additionally assimilated, including Image Analyses, daily and weekly ice charts, as well as Canadian inland lake weekly ice information [7]. The Charts experiment represents the currently operational version of the RIOPS ice analysis system. In the SAR1, SAR1.5, and SAR2 experiments, the three types of SAR retrievals described above are assimilated. In the SAR2C experiment, both SAR retrievals and manual CIS products are assimilated. In each experiment, the 3-D variational (3-D Var) data assimilation system [16] is used to produce gridded ice concentration analyses every 6 h. At each analysis time, the ice concentration field derived at the previous time serves as the background state.

In the SAR1 experiment, the ice concentration threshold for background check (0.85) is implemented as follows. In the case where the SAR retrieval is ice, the threshold value is assimilated if the corresponding background ice concentration is lower than the threshold value; otherwise, the ice SAR retrieval is not assimilated.

The observation-error standard deviation is selected to be 0.1 for the SAR retrievals at every 5 km (SAR1) which is the same value used for assimilating the CIS ice charts in the Charts experiment. For the SAR retrievals at every 2.05 km (SAR1.5), the observation-error standard deviation is estimated to be 0.24, which is 0.1 times the square root of the ratio between the number of low-resolution (5 km) retrievals and the number of high-resolution (2.05 km) retrievals. For the SAR2 and SAR2C experiments, the observation-error standard deviation σ is defined as follows:

$$\sigma = \left[\frac{\sigma_0^2 - \sigma_1^2}{N_{\max} - N_{\min}}(N - N_{\max}) + \sigma_0^2\right]^{0.5}, \qquad (2)$$

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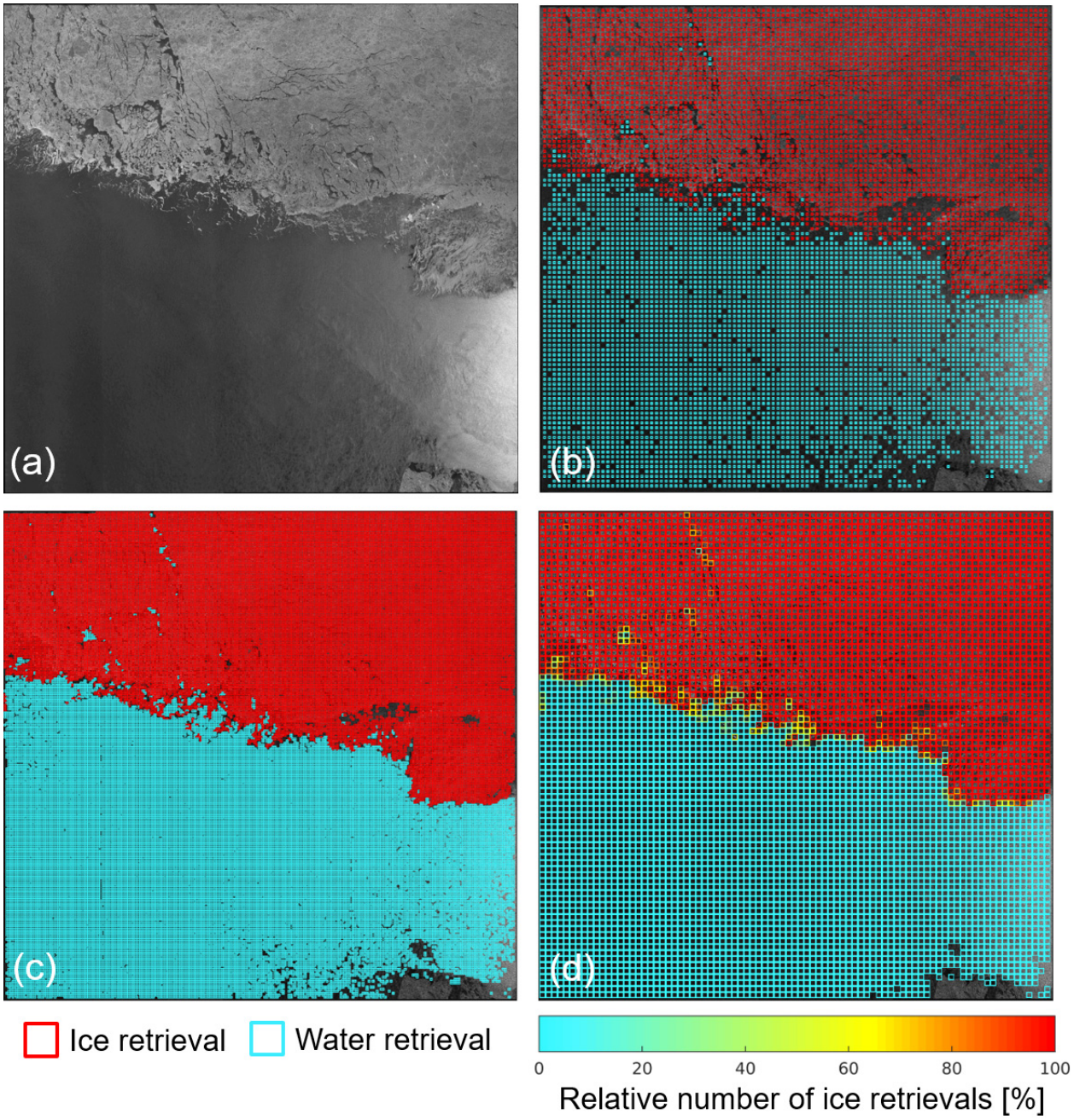


Fig. 6. Example of different types of SAR retrievals. (a) HH channel of RADARSAT-2 HH-HV image (500 km × 500 km) acquired on September 22, 2013 over the Chukchi Sea. (b) Ice (red) and water (blue) 2.05 km × 2.05 km retrievals at every 5 km. (c) Ice/water retrievals at every 2.05 km. (d) Relative number of ice retrievals within 6.15 km × 6.15 km areas at every 6.15 km.

where N is the number of ice/water retrievals used for estimating the relative number of ice retrievals within a 6.15 km × 6.15 km box, $N_{\min} = 1$ is the minimum number of ice/water retrievals possible, and $N_{\max}$ is the maximum number of ice/water retrievals possible (varying from 1 to 9). $N_{\max}$ could potentially be smaller than 9 because only locations that are completely free of land are included. σ_0^2 is the observation error variance in the case where all possible ice/water retrievals are present. At every 5 km, $\sigma_0 = 0.1$; therefore, for 6.15 km spacing, σ_0 is proportionally set to 0.08. σ_1^2 is the observation error variance in the case of assimilating ice/water retrievals at every 2.05 km, $\sigma_1 = 0.24$. Thus, for the SAR2 and SAR2C experiments, the observation-error standard deviation varies between 0.08 and 0.24. The lowest value (giving the observation a relatively larger influence on the ice concentration analysis) is assigned when all possible ice/water retrievals are present ($N = N_{\max}$), and the largest value (giving the observation a smaller influence) is assigned when only a single retrieval is present ($N = N_{\min}$).

IV. Verification of Ice Analyses

In this section, we present verification results for the gridded ice concentration analyses produced by our data assimilation

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TABLE I
SUMMARY OF SAR DATA ASSIMILATION EXPERIMENTS

Experiment	Description	Ice concentration threshold in background check for SAR retrievals	SAR observation error standard deviation
Control	All available satellite observations, but no SAR information	N/A	N/A
Charts	Same as Control, but CIS Image Analysis products included	No threshold	0.1
SAR1	Same as Control, but SAR retrievals conducted at every 5 km (Fig. 6b) are included	0.85	0.1
SAR1.5	Same as Control, but SAR retrievals conducted at every 2.05 km (Fig. 6c) are included	No threshold	0.24
SAR2	Same as Control, but SAR relative number of ice retrievals within 6.15 km × 6.15 km areas (Fig. 6d) are included	No threshold	0.08 – 0.24
SAR2C	Same as SAR2, but CIS Image Analysis products are included.	No threshold	0.08 – 0.24

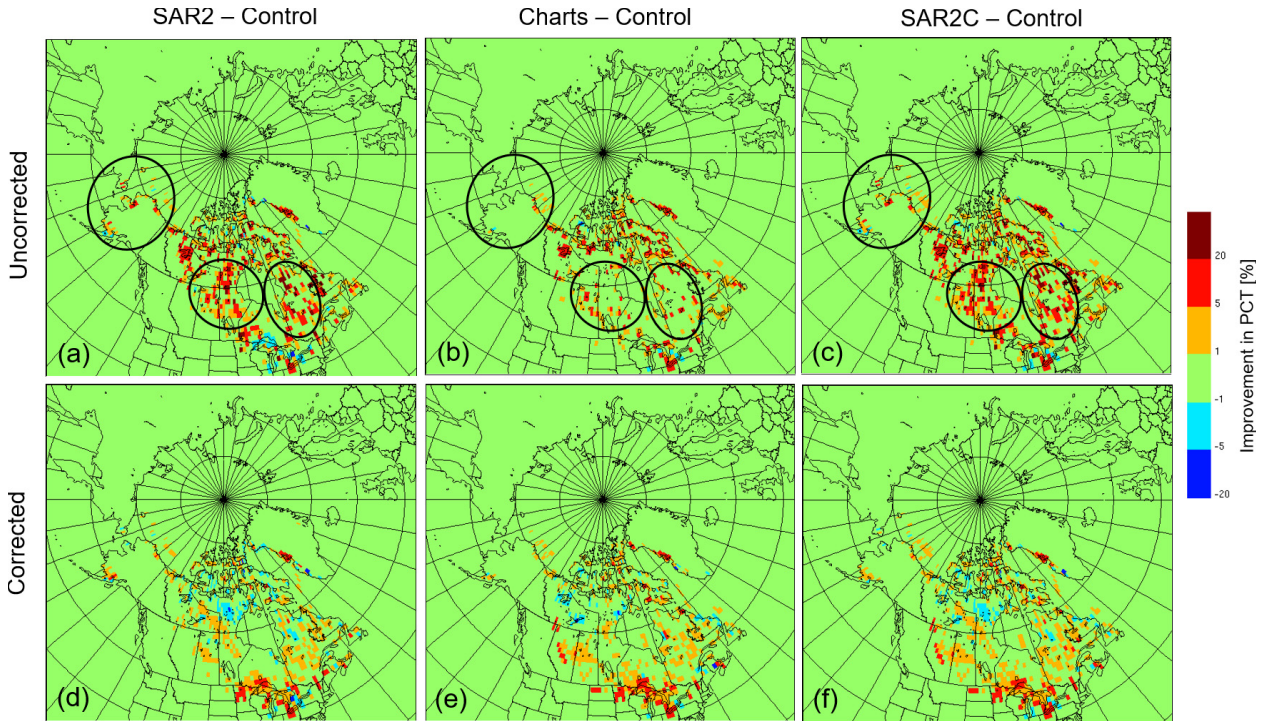


Fig. 7. Maps of improvements in PCT with respect to IMS for the entire year 2013. (a) SAR2 experiment, uncorrected field. (b) Charts experiment, uncorrected field. (c) SAR2C experiment, uncorrected field. (d) SAR2 experiment, corrected field. (e) Charts experiment, corrected field. (f) SAR2C experiment, corrected field. Circled areas in (a) and (c) highlight improvements in ice analysis near coast and over lakes when SAR retrievals are introduced compared to (b) without SAR retrievals.

experiments. Since there is not a perfectly accurate global ice data source, the following four verification data sets, each with their own limitations, were chosen: 1) U.S. Interactive Multisensor Snow and Ice Mapping System (IMS) data [17]–[19], representing daily ice/water products over the entire Arctic with the spatial resolution of 4 km; 2) CIS regional weekly charts [7] over several regions covering the Canadian Arctic and adjacent waters; 3) NIC bi-weekly charts covering the entire Arctic [14]; and 4) CIS Image Analysis products [7].

A. Verification Against IMS

For each data assimilation experiment, we calculated the spatial distribution of percentage correct total (PCT) between

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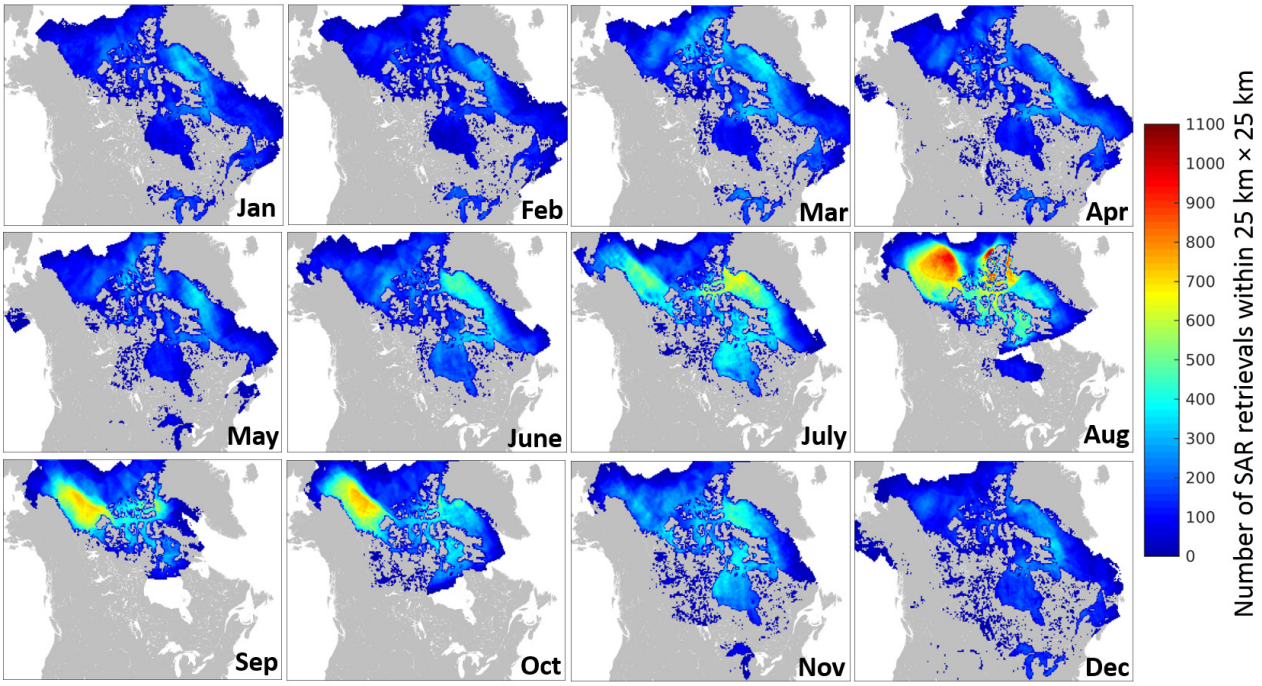


Fig. 8. Distribution of SAR retrievals (in SAR2 experiment) in each month of 2013 calculated in 25 km × 25 km boxes.

the ice concentration analyses (using a threshold of 0.25 to convert ice concentrations to ice/water) and IMS ice/water products in 1° × 1° boxes over the entire year 2013. To assess the impact of SAR-derived information on resulting ice analyses, the difference in PCT between each experiment and control experiment was calculated. In our evaluation, we considered both the uncorrected and corrected ice concentration fields [6]. The corrected field represents an output from a postprocessing step applied to the uncorrected field (which is the ice concentration field propagated through time within the 3-D Var data assimilation cycle). The correction procedure has no effect in areas where the estimated analysis-error standard deviation is below a specified threshold (set to 0.6). In situations where the analysis-error standard deviation is above that threshold, the ice concentration at that location is replaced by an interpolation of the values from the nearby regions where the analysis-error standard deviation is below the threshold.

Fig. 7 shows spatial distributions of PCT difference for SAR2, Charts, and SAR2C experiments for uncorrected [see Fig. 7(a)–(c)] and corrected [see Fig. 7(d)–(f)] ice concentration fields. In the SAR2 experiment, the automatically derived SAR retrievals are assimilated, but manually derived ice information from SAR are not. In contrast, in the Charts experiment, CIS products are assimilated, but automatically derived SAR retrievals are not. The SAR2C experiment combines both the automatically derived SAR retrievals and the manually produced CIS products. From Fig. 7(a), it is seen that the introduction of SAR retrievals in the data assimilation system leads to significant improvements in the ice concentration analysis in the vicinity of land and over inland lakes. This result is explained by the fact that other satellite observations with relatively coarse spatial resolution (such as passive microwave) could not be assimilated in such situations. By comparing Fig. 7(a) and (b), one may observe that the magnitude of improvements in the SAR2 experiment is greater than that in the Charts experiment over the circled geographical areas. Also, the spatial distribution of improvements in PCT in the SAR2 experiment resembles the distribution of improvements in the Charts experiment. The reason for this is that a large portion of the SAR images used for automated retrievals were also used for the manual production of CIS ice charts. However, we note that occasionally some of the

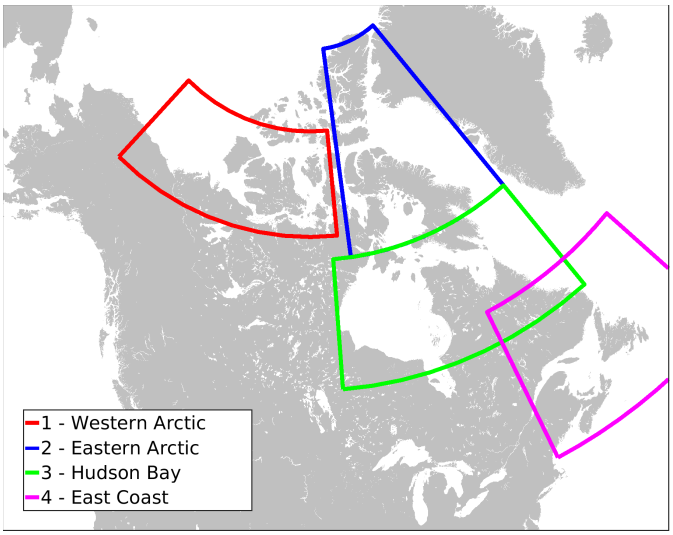


Fig. 9. Four regions selected for verification against CIS and NIC charts covered by both data sources.

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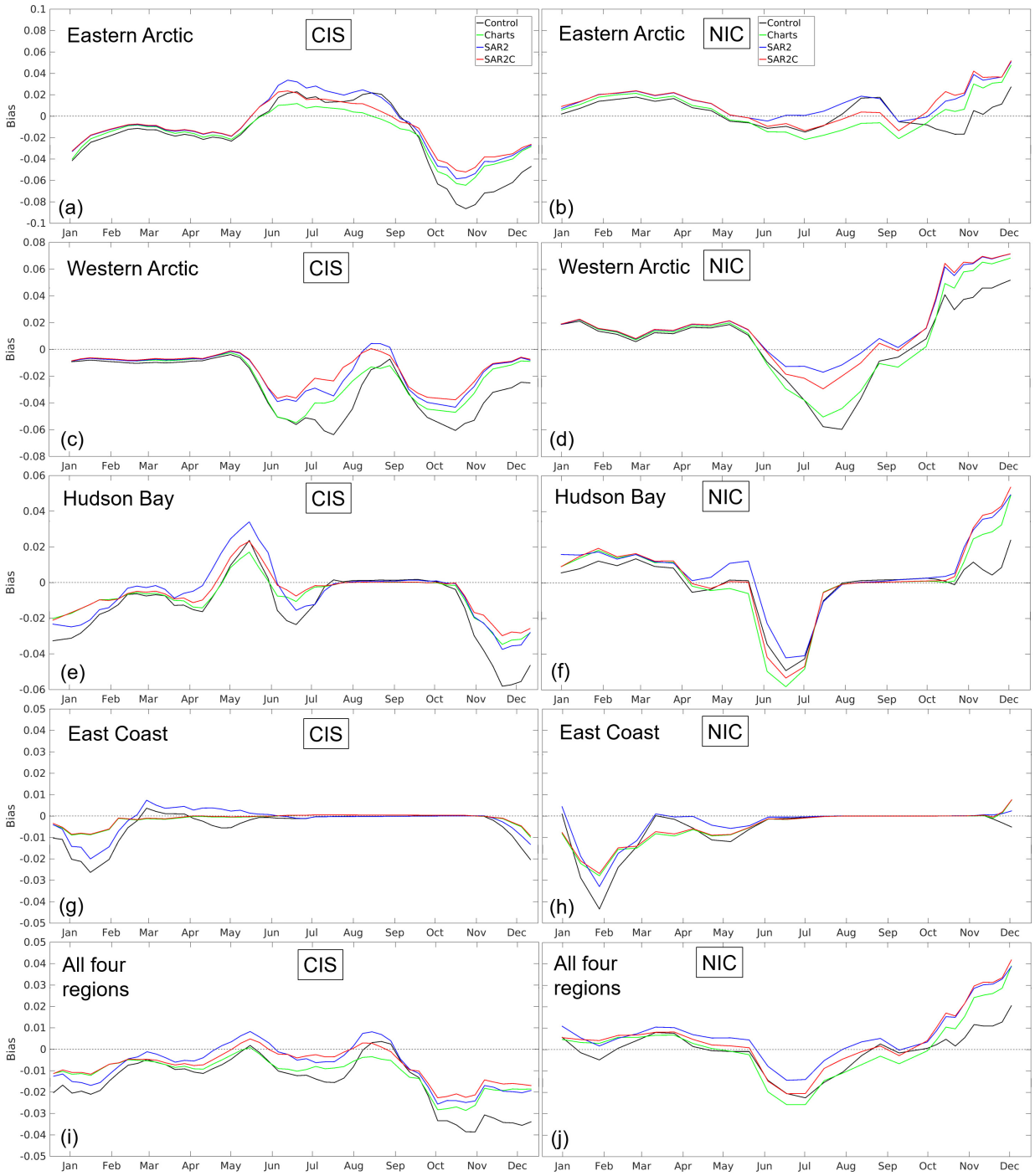


Fig. 10. Bias versus time (year 2013) for verification against CIS and NIC weekly charts for different regions. (a) CIS, Eastern Arctic. (b) NIC, Eastern Arctic. (c) CIS, Western Arctic. (d) NIC, Western Arctic. (e) CIS, Hudson Bay. (f) NIC, Hudson Bay. (g) CIS, East Coast. (h) NIC, East Coast. (i) CIS, all four regions. (j) NIC, all four regions. Color lines correspond to different data assimilation experiments summarized in Table I.

CIS products could be produced from RADARSAT-2 images with other polarization options (such as VV-VH) or other image acquisition modes (such as Ocean Surveillance, Very wide swath, Near incidence angle) or based on complementary images acquired by other satellite sensors such as optical MODIS and VIIRS. Therefore, the combination of automated SAR retrievals and manual CIS products in SAR2C experiment leads to the best result as shown in Fig. 7(c). In the case of the corrected ice concentration field, similar results are observed, but with a smaller magnitude [see Fig. 7(d)–(f)].

To further evaluate the impact of SAR retrievals on the resulting ice concentration analyses, the spatial distribution of SAR retrievals (in the case of the SAR2 experiment) in 25 km × 25 km boxes for each month was computed (Fig. 8). As expected, the spatial distribution of SAR retrievals follows a typical CIS annual RADARSAT-2 image acquisition

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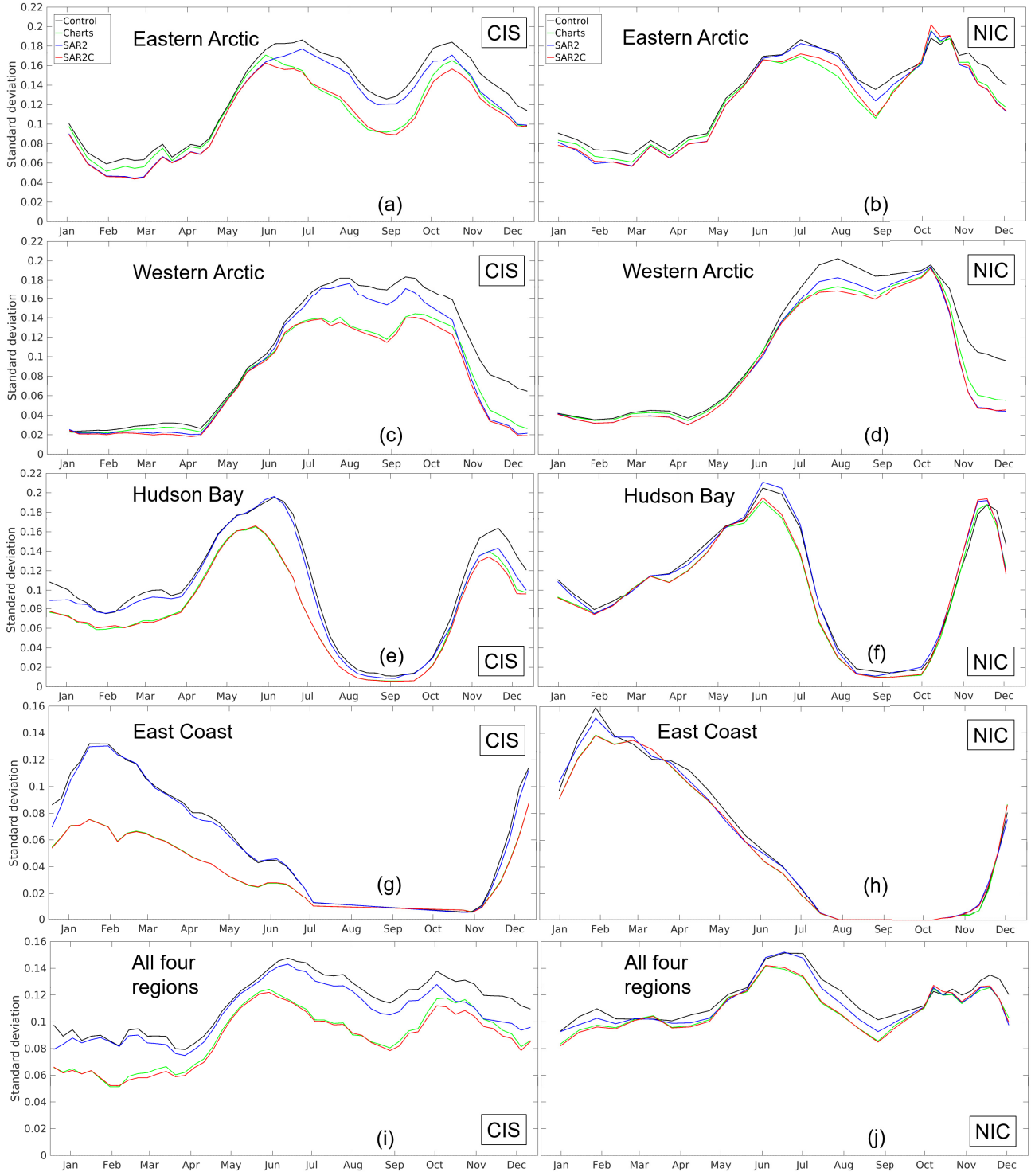


Fig. 11. Standard deviation versus time (year 2013) for verification against CIS and NIC weekly charts for different regions. (a) CIS, Eastern Arctic. (b) NIC, Eastern Arctic. (c) CIS, Western Arctic. (d) NIC, Western Arctic. (e) CIS, Hudson Bay. (f) NIC, Hudson Bay. (g) CIS, East Coast. (h) NIC, East Coast. (i) CIS, all four regions. (j) NIC, all four regions. Color lines correspond to different data assimilation experiments summarized in Table I.

plan designed to support the CIS mandate. During winter months, a homogeneous SAR coverage of the Canadian Arctic and adjacent waters is observed. During summer months, a large number of SAR observations are derived over the northern regions, particularly over the Beaufort Sea, which is associated with a large number of SAR images acquired by CIS in support of summer shipping and navigation activities. Maps of PCT improvements shown in Fig. 7 reflect the SAR data coverage. Despite the fact that a relatively small number of SAR observations were present over the Canadian inland lakes (as shown in Fig. 8), these SAR retrievals translated into a large improvement in ice concentration analyses as

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TABLE II
SUMMARY OF VERIFICATION RESULTS AGAINST CIS REGIONAL CHARTS FOR DIFFERENT REGIONS

Experiment	Eastern Arctic		Western Arctic		Hudson Bay		East Coast		All 4 regions	
	Bias	Std	Bias	Std	Bias	Std	Bias	Std	Bias	Std
Control	-0.0193	0.1424	-0.0271	0.1201	-0.0124	0.1200	-0.0057	0.0836	-0.0149	0.1171
Charts	-0.0174	0.1217	-0.0210	0.0957	-0.0074	0.0954	-0.0021	0.0503	-0.0107	0.0927
SAR1	-0.0184	0.1296	-0.0252	0.1114	-0.0110	0.1133	-0.0040	0.0806	-0.0134	0.1093
SAR1.5	-0.0095	0.1309	-0.0155	0.1107	-0.0055	0.1145	-0.0020	0.0819	-0.0073	0.1100
SAR2	-0.0096	0.1304	-0.0158	0.1096	-0.0055	0.1135	-0.0021	0.0813	-0.0074	0.1092
SAR2C	-0.0111	0.1173	-0.0142	0.0924	-0.0053	0.0942	-0.0019	0.0502	-0.0073	0.0903

TABLE III
SUMMARY OF VERIFICATION RESULTS AGAINST NIC CHARTS FOR DIFFERENT REGIONS

Experiment	Eastern Arctic		Western Arctic		Hudson Bay		East Coast		All 4 regions	
	Bias	Std	Bias	Std	Bias	Std	Bias	Std	Bias	Std
Control	0.0009	0.1510	0.0164	0.1375	0.0009	0.1295	-0.0050	0.0725	0.0019	0.1217
Charts	0.0085	0.1434	0.0235	0.1246	0.0053	0.1241	-0.0038	0.0680	0.0064	0.1148
SAR1	0.0079	0.1472	0.0235	0.1264	0.0053	0.1312	-0.0040	0.0707	0.0062	0.1188
SAR1.5	0.0178	0.1466	0.0328	0.1222	0.0109	0.1334	-0.0027	0.0707	0.0120	0.1189
SAR2	0.0168	0.1445	0.0313	0.1221	0.0104	0.1313	-0.0028	0.0700	0.0118	0.1174
SAR2C	0.0169	0.1442	0.0316	0.1200	0.0088	0.1276	-0.0035	0.0679	0.0108	0.1155

demonstrated in Fig. 7. Therefore, in situations where no other observations could be assimilated (e.g., passive microwave owing to their low spatial resolution), the assimilation of even a relatively small number of SAR observations could result in a considerable improvement in ice concentration analyses.

B. Verification Against CIS and NIC Charts

To further verify the resulting ice analyses produced by different data assimilation experiments, we use the CIS and NIC charts. Ice charts are comprised of a set of image polygons manually produced by an ice analyst. A single ice concentration value is assigned to each polygon that is assumed to be homogeneous. At the same time, the assimilation of SAR retrievals likely increases the ice concentration spatial variability by resolving ice features from one to two orders of magnitude smaller than polygons. This would penalize the standard deviation scores from the experiments assimilating the SAR retrievals.

Four regions illustrated in Fig. 9 were chosen. These regions are covered by both the CIS weekly and NIC bi-weekly charts. For each region, we calculated the bias and standard deviation of the difference as functions of time (year 2013). Figs. 10 and 11 show the bias and standard deviation results (smoothed in time by applying the moving average filter), respectively, against the CIS and NIC charts for four selected data assimilation experiments (Control, Charts, SAR2, and SAR2C). Tables II and III show the average bias and standard deviation values over the entire year for each experiment and each region for the CIS and NIC charts, respectively. From Fig. 10, one may observe that the introduction of SAR-derived information tends to improve bias, particularly in Western and Eastern Arctic during summer time. It is also seen that for the same regions during winter time, bias tends to be

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TABLE IV
SUMMARY OF VERIFICATION RESULTS AGAINST CIS IMAGE ANALYSES FOR DIFFERENT REGIONS

Experiment	Eastern Arctic		Western Arctic		Hudson Bay		East Coast		All 4 regions	
	Bias	Std	Bias	Std	Bias	Std	Bias	Std	Bias	Std
Control	0.0003	0.2134	-0.0070	0.2412	-0.0155	0.1647	-0.0153	0.1738	-0.0117	0.1837
Charts	-0.0129	0.1700	-0.0128	0.1979	-0.0098	0.1497	-0.0090	0.1611	-0.0104	0.1608
SAR1	-0.0036	0.1965	-0.0074	0.2232	-0.0111	0.1601	-0.0111	0.1724	-0.0094	0.1761
SAR1.5	0.0092	0.2040	0.0127	0.2279	-0.0050	0.1639	-0.0043	0.1756	-0.0008	0.1807
SAR2	0.0078	0.2026	0.0108	0.2270	-0.0057	0.1624	-0.0045	0.1744	-0.0016	0.1794
SAR2C	-0.0067	0.1717	-0.0044	0.1972	-0.0076	0.1492	-0.0079	0.1606	-0.0073	0.1607

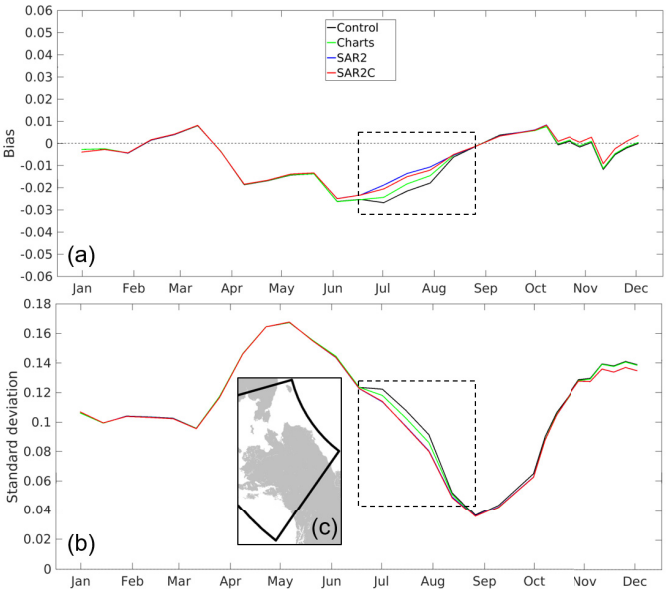


Fig. 12. Verification results against NIC charts for the Alaska region. (a) Bias. (b) Standard deviation. (c) Alaska region. Color lines in (a) and (b) correspond to different data assimilation experiments summarized in Table I. Dashed rectangles indicate time periods where SAR data provide substantial improvement in the ice analysis.

negative when compared with CIS charts, and it tends to be positive when compared with NIC charts. Overall, with the introduction of SAR data, bias tends to improve when compared with CIS charts, and it tends to slightly degrade when compared with NIC charts (Fig. 10; Tables II and III); this can be attributed to the different nature of the CIS and NIC chart production. However, with the introduction of SAR data, standard deviation improves when compared with both CIS and NIC charts (Fig. 11; Tables II and III). Fig. 11 shows that the standard deviation is most prominently improved in the Charts and SAR2C experiments when compared with CIS charts and less improved when compared with NIC charts. This is likely because the Charts and SAR2C experiments include the assimilation of the CIS Image Analysis products that contribute to the production of the weekly chart used for verification. From that point of view, NIC charts are a more independent source of data verification than CIS charts. On the other hand, CIS charts are produced more frequently and only for the CIS regions of interest, whereas NIC charts are produced for the entire Arctic domain.

We also found that SAR1.5 and SAR2 result in very similar bias and standard deviation (not shown). However, SAR2 uses a much smaller number of SAR observations (as it takes the 3×3 average of ice/water retrievals) than SAR1.5, which makes the assimilation of SAR observations more efficient. Overall, for all regions, among SAR1, SAR1.5, and SAR2, the best verification scores are observed for the SAR2 experiment. The use of both SAR retrievals and charts in the SAR2C experiment leads to the best verification scores among all experiments (Tables II and III).

We note that, over the selected four regions, CIS products are available on a regular basis, and, therefore, they can be used for data assimilation; however, over the regions where CIS products are available less frequently or not available (such as inland lakes), SAR retrievals provide a significant positive impact on the resulting ice analyses (as shown in Section IV-A). To further demonstrate this, we selected an area surrounding Alaska and calculated bias and standard deviation with respect to NIC charts specifically over this area (Fig. 12). One may observe that improvement in bias and standard deviation during summer time is larger in the case of assimilating SAR retrievals than in the case of assimilating CIS charts (the Charts experiment). This is attributed to the fact that the coverage of SAR data is more extensive than that of CIS charts over that region.

C. Verification Against CIS Image Analyses

The valid times of the CIS regional charts used for verification presented in Section IV-B correspond to the valid times

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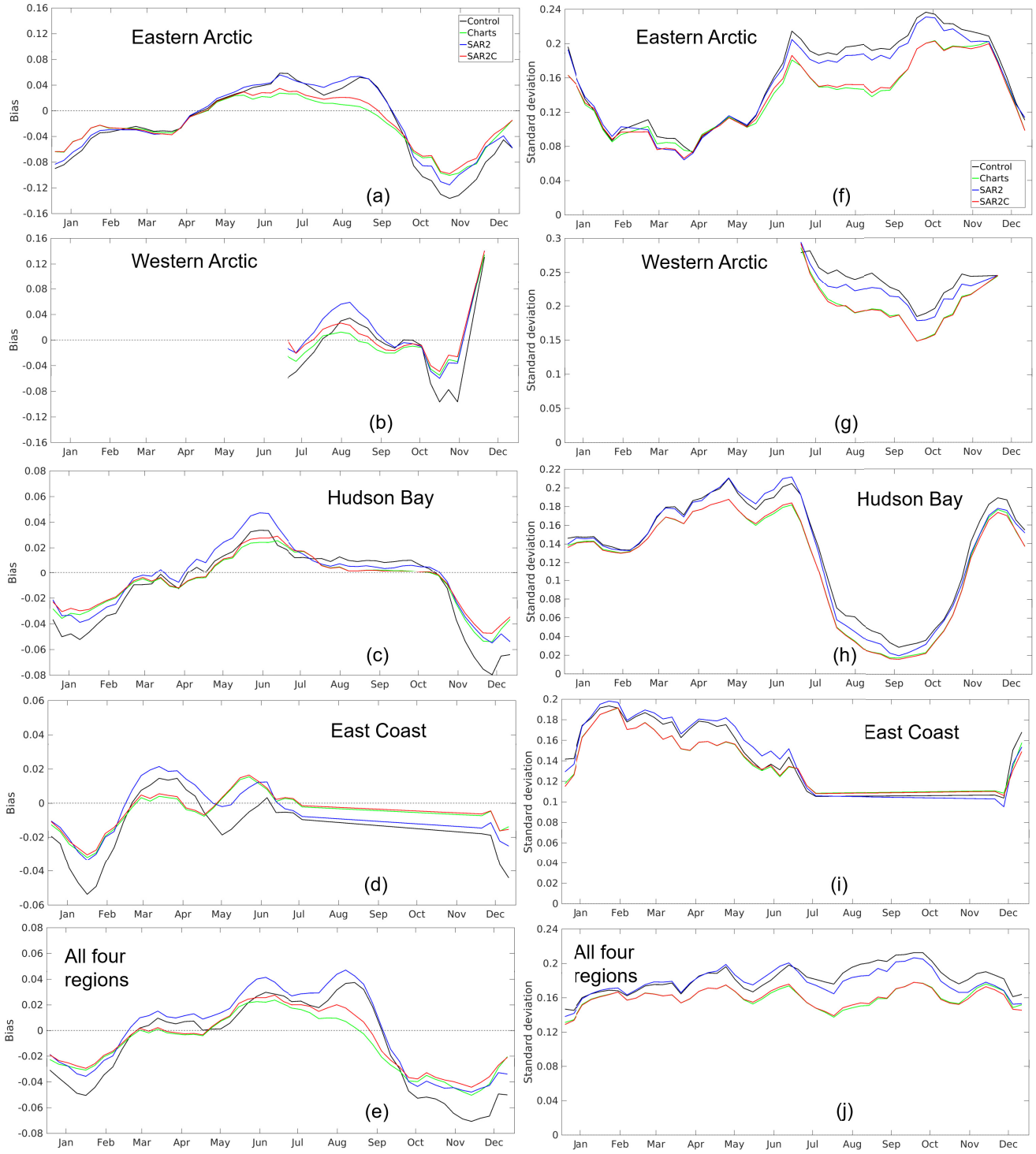


Fig. 13. Verification results against CIS Image Analyses for four regions. (a)–(e) Bias. (f)–(j) Standard Deviation. Color lines correspond to different data assimilation experiments summarized in Table I.

of the ice analyses products (1800 UTC of each Monday). However, individual Image Analysis products contributing to the creation of a given regional chart are generated from SAR images acquired throughout a one-week period surrounding the valid time (1800 UTC Monday). The Image Analysis products with valid times preceding 1800 UTC Monday are assimilated as part of the Charts and SAR2C experiments. Therefore, the verification of ice analyses produced by Charts and SAR2C experiments against CIS regional charts is not completely independent. In the RIOPS data assimilation system, CIS regional charts are assimilated only where the analysis-error standard deviation is greater than 0.4. Therefore, potentially less of the CIS regional charts are assimilated when SAR data is directly assimilated.

In this section, we present the verification of 3-D Var ice concentration analyses against CIS Image Analysis products

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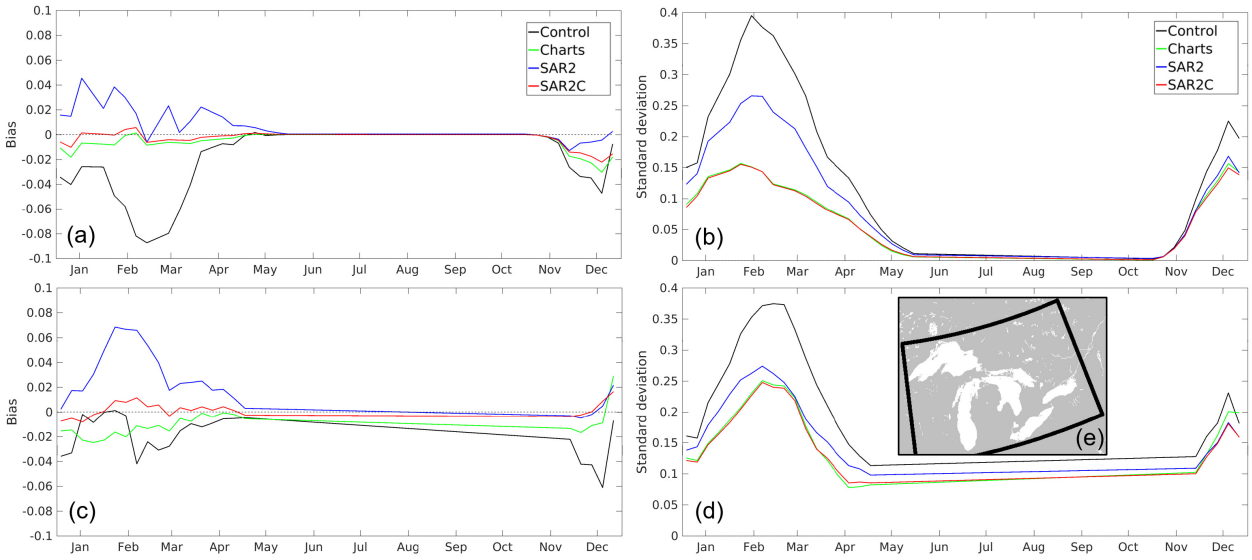


Fig. 14. Verification results against CIS regional charts and Image Analyses for the Great Lakes region. (a) and (b) Bias and standard deviation for verification against CIS regional charts. (c) and (d) Bias and standard deviation for verification against CIS Image Analyses. (e) Great Lakes region. Color lines correspond to different data assimilation experiments summarized in Table I.

that have not yet been assimilated. Ice analyses valid at 0600 UTC are verified against all Image Analyses with valid times between 0900 and 1500 UTC. To be consistent with Section IV-B, this verification is conducted for the same four regions presented in Fig. 9. Fig. 13 shows bias and standard deviations as the functions of time for these regions. Table IV shows average verification scores for different regions. It is seen that for all regions, the introduction of SAR retrievals generally leads to improvement in standard deviation compared to the control experiment with bias staying at a low level.

One limitation of this verification exercise is that the considered regions might not be completely covered by the CIS Image Analyses products. Furthermore, the number of observation points used for calculating verification scores varies throughout the year. In May–June, the standard deviation of SAR2 is slightly higher than that of the Control experiment (Fig. 13), but the number of points used for calculating the verification scores was the smallest for this time period.

From Fig. 13, one may observe that the drop in the standard deviation of Charts and SAR2C experiments with respect to the Control run is much smaller than the situation where verification was conducted against CIS regional charts (Fig. 11). This effect is also well noticeable in Fig. 14, showing verification results for the Great Lakes region in the case of using CIS regional charts and CIS Image Analyses products. Note that the Great Lakes are covered by CIS charts, but not by NIC charts.

V. Conclusion

SAR data assimilation in sea ice numerical prediction systems is a relatively new area of research. Despite the fact that operational spaceborne SAR observations in Canada over the Arctic have been available since 1995 with the launch of RADARSAT-1, the automated use of SAR data in numerical prediction systems has not been feasible until recently.

In this article, we investigated how ice and water retrievals automatically derived from RADARSAT-2 HH-HV SCW images impact ice concentration analyses at different regions and in different months. To this end, we derived ice and water information from more than 7000 SAR images acquired over the Canadian Arctic and adjacent waters in 2013 using our previously introduced approach [9], [10]. An additional quality-control procedure was developed and applied to the derived ice and water retrievals.

We designed a series of data assimilation experiments with and without including SAR-derived information over the entire year 2013. Since there is no perfectly accurate ice data set with a global coverage, we chose four different data sources for verification: IMS, CIS Regional Charts, NIC Arctic Charts, and CIS Image Analyses. Our verification results against these ice data sources suggest that the inclusion of automatically derived ice and water retrievals in the data assimilation cycles leads to improvements of ice concentration analyses in the situations where the high spatial resolution is important (e.g., near shoreline and over small inland lakes). Automatically derived SAR retrievals are particularly useful over the areas where the CIS Image Analysis products are not available or have limited coverage.

The assimilation of the SAR ice and water retrievals at a high spatial resolution increases the spatial variability of the ice concentration analyses. We are developing additional verification methods that will not unfairly penalize the increase in the spatial variability of the ice concentration field, as is now likely happening when using CIS and NIC ice charts for verification. Also, the RIOPS ice data assimilation system is designed to estimate the value of the continuous ice concentration variable. Therefore, the use of binary ice/water retrievals in the system is not ideal. Future research should be focused on more accurate retrieval of ice concentration from SAR.

The automated ice and water retrieval techniques will be adapted to the upcoming data stream from the three-satellite

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RCM. With the increased spatial and temporal coverage of RCM, the assimilation of automatically derived ice and water information from SAR data in RIOPS should further enhance ice concentration analyses.

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